MAJOR PROJECT SYNOPSIS · PHASE I PROPOSAL · B.TECH COMPUTER SCIENCE & ENGINEERING · PROJECT TEAM ID CSE27-293

MFSTNet — Multimodal Spatio-Temporal Congestion Forecasting for Adaptive Traffic Signal Control

working name: cnn-vit-bilstm-cross-attention-traffic-system

A proposal to forecast per-lane congestion 60 seconds ahead from an existing traffic camera alone, and to let a reinforcement-learning controller act on that forecast — no new roadside sensor, and no raw frame leaving the camera.

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What this proposal covers

The deck follows the synopsis chapter order, so a panel reading along can follow.

- **1 Introduction**
  Background, motivation, and how the idea was selected from five
- **2 Literature Review**
  Four strands, each mature on its own, and the gap between them
- **3 Problem Statement & Objectives**
  One problem, three research questions, six objectives O1-O6
- **4 Proposed Methodology**
  Design decisions, architecture, corpus, model, control, plan
- **5 Hardware & Software**
  What the project needs — one laptop and one 6 GB GPU
- **6 Expected Outcomes**
  Seven deliverables and the targets set in advance for each
- **7 Conclusion**
  What is committed to, and what is deliberately left out
- **References**
  IEEE style, 16 entries, each cited in the report

One thing to hold onto throughout: this is a proposal. Every number in it is a target set in advance, not a measurement. Nothing here reports a result, because the work starts now.

A signal allocates green from a survey taken once

Fixed-time control computes cycle length and phase splits from a manual count, then holds them for years.

- **The plan is right on average**
  and therefore wrong most of the time — green is held on an empty approach while a saturated one waits.
- **The stream is heterogeneous**
  A two-wheeler, an auto-rickshaw and a bus occupy different road space and discharge at different rates.
- **The stream is not lane disciplined**
  Vehicles sit between nominal lanes, so queue geometry does not follow the markings on the road.
- **Counting vehicles measures the wrong thing**
  Indian practice already says so: the capacity manual publishes passenger-car equivalents for exactly this reason.

WHY THIS FRAMING

- **Cameras already exist**
  installed for enforcement, at most junctions
- **Loops and probes do not**
  and cost money the deployment does not have
- **A decision acts on the future**
  the next 30-90 s, not the present queue
- **So forecast, do not measure**
  one cycle ahead is the useful horizon

Camera-only is a harder problem than sensor-fusion. It is chosen because it is the one that can actually be deployed on the junctions this targets.

How the idea was chosen

Five candidates were scored on four criteria before one was fixed. The shortlist is recorded rather than discarded.

CANDIDATE IDEA	DECISION
Helmet and traffic-violation detection from CCTV	Rejected — no forecasting or control component, so the project would end at a detector and have nothing to ablate.
Vehicle re-identification across junctions	Rejected — the footage the team can obtain is single-view and unsynchronised, so the core assumption fails before any modelling.
Accident and anomaly detection	Rejected — the positive class would be too small to evaluate honestly within a twenty-week schedule.
Sensor-fusion congestion forecasting (camera + loop detector + GPS)	Rejected — no route to the loop-detector or probe-vehicle data; the project would stall on a dependency outside the team's control.
Camera-only congestion forecasting driving an adaptive signal controller	Selected — every input is obtainable from a single fixed camera, and the work splits cleanly into a perception claim and a control claim that can each stand alone.

The four criteria: every input obtainable without a dependency outside the team; a component that can be ablated and therefore defended; a fit inside twenty weeks at a zero-rupee cash budget; and a negative result that is still a reportable finding rather than a failed project.

Four strands, each mature on its own

Every mechanism this project uses is published. What is not published is the combination, for this setting, reported this way.

- **Visual representation [2] [3] [4]**
  Residual networks are strong locally; a queue is a global quantity. Self-supervised ViT features transfer without fine-tuning — which is what makes a frozen branch affordable.
- **Temporal modelling [14] [15]**
  Sixty steps is short: long enough for a recurrence to carry something, too short to train a transformer from scratch on a corpus this size.
- **Fusion [1]**
  Cross-attention and gated combination are standard. What is never reported is the gate itself — the quantity that says which branch the model actually relied on.
- **Signal control [7] [9] [10] [13]**
  PPO on a SUMO intersection is established. The recurring methodological problem in the surveys is weak baselines and single-seed reporting.

THE GAP, IN ONE SENTENCE

No fixed-camera junction dataset for unstructured traffic carries per-lane congestion labels; no published work reports what a CNN-to-ViT fusion gate learns on traffic imagery; and learned controllers are rarely measured against a baseline that was itself tuned first.

One problem, stated precisely

Every clause below is load-bearing; each one excludes an easier problem we are choosing not to solve.

> Given only the video streams of fixed cameras at a signalised junction carrying heterogeneous, non-lane-disciplined traffic, predict the congestion level of each approach 60 seconds ahead, and use that prediction to select the next phase and its duration so that mean waiting time is lower than under a tuned fixed-time plan — with no raw frame transmitted off the capture device.

- **RQ1**
  Do frozen appearance features add anything over vehicle counts alone when forecasting per-lane congestion sixty seconds ahead? The count-only baseline is part of the experiment, not a courtesy.

- **RQ2**
  Does gated bidirectional fusion beat cheaper fusion, and does the learned gate carry an interpretable signal about which branch the model relies on?

- **RQ3**
  Does a controller that receives a short-horizon congestion forecast outperform the same controller without one, measured against a swept-and-tuned Webster baseline rather than an untuned one?

- **Why each is stated this way**
  A negative answer to any of the three is a reportable finding, not a failed project. That is deliberate.

Six objectives, in dependency order

O1-O3 are Phase I and mandatory. O4-O5 are Phase II. O6 is Phase III and optional.

- **O1**
  Assemble a licence-audited corpus of fixed-camera junction footage covering unstructured traffic, and fine-tune a YOLOv8s detector over eight vehicle classes that include the auto-rickshaw, e-rickshaw and cattle absent from COCO.

- **O2**
  Derive per-lane congestion labels from detector counts using PCU-weighted occupancy with per-camera thresholds, so that a label means the same thing at a six-lane arterial and a two-lane approach.

- **O3**
  Implement the Phase I model — two frozen backbones, standard cross-attention fusion, a BiLSTM and a per-lane congestion head — and demonstrate that it trains to convergence before any Phase II component is added.

- **O4**
  Extend the model with gated bidirectional fusion, temporal self-attention and attention pooling, and quantify each addition through an eight-configuration ablation that includes a linear probe as its floor.

- **O5**
  Evaluate leave-one-camera-out, reporting mean and bootstrap confidence interval across folds together with the seed spread, so that no reported score is a memorised camera or a lucky initialisation.

- **O6**
  Train a PPO signal controller on a SUMO intersection whose observation vector carries the forecast, measure it against a swept, best-configuration Webster baseline over thirty paired seeds, and demonstrate an edge-to-server prototype in which no raw frame leaves the capture device.

What we chose, and what we changed

Thirteen decisions were taken from a shortlist. Six of them are where the design moved most.

COMPONENT	CONSIDERED	SELECTED
Vision transformer	Supervised ViT-S/16; CLIP ViT-B/32; DINOv2 ViT-S/14	DINOv2 ViT-S/14
Backbone regime	Full fine-tuning; partial unfreeze partway through training; frozen backbones with a precomputed feature cache	Frozen with a cache
Fusion mechanism	Channel concatenation; one-way cross-attention; bidirectional cross-attention averaged; gated bidirectional cross-attention	Gated bidirectional
Evaluation split	Random 60/20/20; stratified by class; leave-one-camera-out	Leave-one-camera-out
Edge device	NVIDIA Jetson Nano; Raspberry Pi with a Coral accelerator; a team laptop	Team laptop
Datastore	PostgreSQL with TimescaleDB; InfluxDB; SQLite	SQLite

TWO OF THESE ARE FORCED, NOT PREFERRED

Frozen backbones are forced by 6 GB of video memory — an uncached step at the specified batch size pushes 1,920 frames through both backbones and does not fit. The edge device is a team laptop because the cash budget is zero, which is why every latency figure this project reports must name the host it was measured on.

Three tiers, one privacy boundary

Frames are read, processed and discarded inside the edge tier. Only counts, predictions and commands cross the network.

EDGE · TEAM LAPTOP AT THE CAMERA

- Fixed CCTV frames — never transmitted, never written to disk
- YOLOv8s detection over 8 India-relevant classes
- Per-lane counting on hand-placed, reviewed lane centres
- Local Webster fallback, armed if the broker is silent > 10 s

SERVER · FASTAPI + ONNX + PPO

- MFSTNet (ONNX) — per-lane congestion, 60 s ahead
- PPO policy — 16-dim observation, 12 discrete actions
- SignalGuard — min/max green, all-red, clamping outside HTTP
- SQLite — bounded history of counts and decisions

DASHBOARD · REACT

- Live panel — counts, predictions, gate value
- Benchmark panel — from the committed CSV
- The two panels never merge
- Refresh budget 2 s

MQTT CONTRACT — EMERGENCY QoS 2 | COUNTS AND COMMANDS QoS 1 | PREDICTIONS AND HEARTBEAT QoS 0

Two fallback paths are required behaviour rather than robustness extras: if the forecast is unavailable the controller runs on counts alone with the four forecast entries zeroed — never with a shortened observation vector — and if the broker is silent for more than ten seconds the edge node arms its own fixed-time controller.

The corpus has to be built

No public dataset gives fixed-camera junction footage with per-lane congestion labels, so the pipeline is part of the work.

PIPELINE, IN ORDER

- **Source and triage**
  curated public footage + campus capture
- **Lane geometry**
  placed by hand, then reviewed — never derived
- **Detection**
  fine-tuned YOLOv8s, per-frame per-lane counts
- **Counts file**
  threshold-free, so relabelling costs seconds
- **Labelling**
  PCU-weighted occupancy, per-camera p33/p67
- **Audit**
  fails loudly; nothing proceeds past it
- **Split**
  leave-one-camera-out, cut by clip — never by window

- **Labels are PCU-weighted, not counts**
  Ten two-wheelers and ten buses are not the same congestion. The capacity manual publishes the equivalences; an unmapped class raises rather than defaulting to 1.0.
- **Lane centres are placed by hand**
  Automatically derived centres move by 0.164 of frame width when the detector changes — silently altering every count the corpus rests on.
- **The split is cut by source clip**
  Overlapping temporal windows from one clip on both sides of a split would leak the answer across it.
- **Two detector tracks run in parallel**
  A public dataset bootstraps week 2 so nothing waits on annotation; every experiment records which weights produced it.

The model, and the gate

Two frozen branches, fused by a learned gate, over a 60-step window at 5 s intervals.

PROPOSED STRUCTURE

- **Frozen backbones**
  `ResNet-50 + DINOv2 ViT-S/14, cached once`
- **Two cross-attention passes**
  `Z_A = CNN queries ViT ; Z_B = ViT queries CNN`
- **Gated fusion**
  `g = sigma(W[Z_A ; Z_B]) ; F = g Z_A + (1-g) Z_B`
- **BiLSTM**
  `2 layers, hidden 128, bidirectional`
- **Temporal self-attention**
  `2 layers, 4 heads, over the BiLSTM outputs`
- **Attention pooling**
  `over time, then per-lane ROI pooling`
- **Per-lane head**
  `LOW / MEDIUM / HIGH for N, S, E, W at t + 60 s`

- **The gate is an output, not a detail**
  It is returned from the forward pass, logged, shown on the dashboard and analysed. A gate that is computed and discarded makes the fusion claim untestable.

- **Frozen is forced, then exploited**
  6 GB cannot hold an uncached step. Frozen features are deterministic, so they are computed once and cached — and the ablation drops from days to hours.

- **Every block is disableable by flag**
  Eight configurations A-H run from configuration alone, with no code edit between them. Config H is a linear probe — the floor.

- **Per-lane read-out, not global pooling**
  Pooling globally and applying one head four times cannot produce four different predictions.

The controller, and the baseline it must beat

PPO on a SUMO four-way intersection, with the forecast inside the observation vector.

- **Observation is 16-dim and a contract**
  Queues, phase, elapsed green, time of day, emergency flag, and four forecast entries. If the forecast is missing those four are zeroed — the vector is never shortened.
- **Twelve discrete actions**
  Two phase groups by six green durations. Reward is the negative mean waiting time across lanes at each decision step.
- **The simulator models the real stream**
  The sublane model is enabled so vehicles occupy lateral positions within a lane; the residual gap is declared, not papered over.
- **Webster is swept, not configured once**
  The published saturation-flow range for Indian conditions spans more than a factor of two. A single arbitrary value produces whichever result the author preferred.
- **The best Webster is the baseline**
  PPO must beat the strongest configuration the sweep finds, over 30 paired seeds across three demand regimes.
- **The forecast has to earn its place**
  A perfect-forecast ceiling run comes first, because it can falsify the whole idea cheaply.

How it will be evaluated

The protocol is fixed now, before any number exists, so that it cannot be chosen afterwards to suit the result.

PROTOCOL, FIXED IN ADVANCE

- **Leave-one-camera-out**
  each fold holds out one camera entirely
- **Five seeds per config**
  a one-seed table ranks initialisations
- **Bootstrap 95% CI**
  percentile method, 10,000 resamples
- **Paired t-test, alpha 0.05**
  with Cohen's d beside it
- **Seed spread reported**
  two configs inside it are not separated
- **Count-only baselines**
  LSTM and XGBoost on the same folds

Raw per-run results are committed, and every table in the final report is generated from them by a committed script — never transcribed.

- **Why hold out whole cameras**
  A random split over windows lets the model recognise a camera it has already seen, and reports that as generalisation.

- **Why five seeds**
  On a corpus this size the movement between seeds can exceed the gap between two architectures. One seed would rank noise.

- **Why count-only baselines are in from week one**
  They answer RQ1 directly. If appearance does not beat counts, that is the finding, and it will be reported as the finding.

Three phases, with gates between them

Nothing from Phase II is written until the Phase I model trains cleanly. This is a constraint, not a preference.

PHASE	WEEKS	CONTENTS	EXIT CONDITION
Phase I Mandatory	Weeks 1-8	Corpus pipeline and audit; YOLOv8s fine-tuning; PCU-weighted labelling; feature cache; CNN + ViT with standard cross-attention, BiLSTM and the per-lane head.	The Phase I model trains to convergence and beats the linear probe on a held-out camera. Nothing from Phase II begins until it does.
Phase II Stretch	Weeks 9-14	Gating replaces standard cross-attention; temporal self-attention over the BiLSTM outputs; attention pooling replaces the last hidden state; the full eight-configuration ablation, leave-one-camera-out at five seeds.	Every configuration runs from a flag with no code edit between configurations, and the ablation table is generated from committed per-run CSVs.
Phase III Optional	Weeks 15-20	SUMO environment and PPO arms; the thirty-seed paired benchmark against a swept Webster baseline; ONNX export; edge node, MQTT contract, control server and dashboard.	The prototype demonstrates both required fallback paths, and the control result is reported with its disqualification rules applied.

MILESTONES

Week 2 Idea fixed, requirements written, detector bootstrapped on a public dataset so the corpus pipeline is not blocked on in-house annotation.

Week 5 Corpus pipeline end to end: triage, reviewed lane geometry, counts, labels, audit. Architecture and design documents due.

Week 8 Phase I model converging; feature cache built; the in-house dataset swaps in and every experiment records which detector weights produced it.

Week 11 Phase II components in; ablation running; test plan and test design documents due.

Week 14 Leave-one-camera-out ablation complete at five seeds, with raw per-run CSVs committed.

Week 17 Controller trained and benchmarked; ONNX export and control server working.

Week 19 Prototype integrated end to end, both fallback paths exercised.

Week 20 Results, test report and paper draft; public code release with pinned requirements and fixed seeds.

What could go wrong, and the response

Eight risks, each with a response rather than a hope. Two of them are outcomes we intend to measure, not avoid.

RISK	LIKELIHOOD	IMPACT	RESPONSE
Annotation effort is underestimated	High	High	Bootstrap the detector on a public dataset from Week 2 so the corpus pipeline never waits on in-house annotation, and curate from permissively licensed sources before collecting anything new.
Permission for campus footage is delayed	Medium	High	The perception track is built on curated public footage first, so in-house capture swaps in as an addition rather than a dependency.
A capture location yields only one camera	Medium	Medium	Leave-one-camera-out cannot score a single-camera location. Capture is planned as multiple viewpoints per location from the start.
The HIGH class is rare in the available footage	High	Medium	Report macro-F1 and per-class support rather than accuracy, weight the loss by inverse frequency, and state the class counts beside every result.
6 GB of VRAM is insufficient	Medium	High	Freeze both backbones and cache their features; hold a small number of windows resident and accumulate gradients to the specified batch size.
The simulator does not reproduce unstructured traffic	High	Medium	Use the sublane model, calibrate demand against observed counts, and declare the residual gap as a limitation rather than claiming realism.
The forecast does not help the controller	Medium	High	Run the perfect-forecast ceiling first: if a perfect forecast does not beat no forecast, the forecast is not worth carrying and that is reported as a finding.
Appearance features do not beat counts	Medium	High	The count-only baseline is in the experiment from the start, and the two claims are stated separately so the control claim ships even if the vision claim does not.

Seven deliverables, with targets set in advance

A target that is missed will be reported as missed, with the measured value — not restated afterwards to match what was obtained.

ASPECT	TARGET	HOW IT IS MEASURED
Detection quality	mAP@0.5 ≥ 0.70 over the eight classes	Measured on a human-labelled test split, never on pseudo-labels.
Detection throughput	≥ 10 frames per second on the edge device	Reported with the measurement host named, because the edge device is a laptop and not a Jetson.
Forecasting quality	Macro-F1 ≥ 0.75 on held-out cameras, and above the count-only baseline	Both halves matter: beating the baseline is the claim, the absolute number is the target.
Model inference	≤ 150 ms per window on server CPU via ONNX	The budget the control loop is designed around.
Control improvement	≥ 15% lower mean waiting time than the best Webster configuration	Measured over thirty paired seeds with a paired t-test and an effect size, and disqualified if fewer than 85% of trips complete.
Decision and message latency	≤ 50 ms per control decision and ≤ 200 ms end to end over MQTT	Measured on the prototype, with the host stated.
Reproducibility	Every table regenerated from committed CSVs by a committed script	Fixed seeds across the numerical, deep-learning and RL libraries.

DELIVERABLES

Curated dataset · Fine-tuned detector · Congestion forecasting model · Ablation and evaluation results · Signal controller · Working prototype · Reproducibility package

What this proposal commits to

Three commitments distinguish this from a description of an architecture.

- **Order**
  Phase I is a mandatory core that must converge before any refinement is written. The gate is a guard against an elaborate model with no experiments attached to it.
- **Reporting the gate**
  The fusion gate is an artefact we publish, not an internal detail. It is what makes the fusion claim checkable at all.
- **Falsifiability**
  A count-only baseline and a linear probe are in the experiment from the start; the evaluation holds out whole cameras; every config runs at five seeds; the classical baseline is tuned before it is beaten.
- **And what is deliberately excluded**
  Multi-junction coordination, trajectory prediction, deployment on live signal hardware, night and adverse weather, and any claim that requires data we do not have.

Whether each component earns its place is a question the experiments are designed to answer honestly, in either direction. The two claims — the count-based forecaster with its controller, and the vision claim — are stated separately so that a null result on one does not take the other down with it.

Sources, IEEE style

16 entries, each cited at first use in the synopsis report.

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PHASE I · PROJECT PROPOSAL

Thank you.
Questions welcome.

CNN VIT BILSTM CROSS-ATTENTION BASED TRAFFIC MANAGEMENT SYSTEM

Project Team ID CSE27-293 · guided by Ms. Neha Pokhriyal, Assistant Professor

working name: cnn-vit-bilstm-cross-attention-traffic-system